Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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Summary

Cloud & DevOps/MLOps Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, delivering DevOps and cloud-platform engagements for financial-services and other enterprise customers. Fluent across AWS landing zones and governance (Organizations, SCPs, Terraform, GitHub Actions), production MLOps on SageMaker (pipelines, blue-green endpoints, drift detection), full-stack apps (React, FastAPI, Node.js), and AI-assisted developer tooling (Claude Code plugins, MCP servers, agent skills). Author of 3 samples published by AWS (aws-samples). 11 upstream PRs merged, including axios, Apache Airflow, lucide, and multer. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems, 107 contests.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Hyderabad, India

- **Lead DevOps Consultant, Enterprise customer (Feb 2026 – Present):** Built a Terraform landing zone: Organizations OUs, Control Tower, SCPs, tag policies, account security baselines and least-privilege deploy roles.
  - Automated account vending with Terraform Cloud and OIDC federation.
  - Automated network onboarding and tag-based firewall access with Transit Gateway.
- **DevOps Consultant, Global financial services customer (Oct 2024 – Aug 2025):** Wrote tested Terraform modules (terraform test, PyTest) for a multi-account landing zone. Worked on preventative security controls and EventBridge-based cross-account automation.
- **DevOps Consultant, Financial market infrastructure customer (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns.
- **MLOps Pipeline (SME program, Jun – Aug 2025):** Built a SageMaker medical-image-classification pipeline that trains, gates and ships models, with blue/green canary endpoints, auto-scaling, and PSI drift detection, provisioned via Terraform + CodePipeline behind CloudFront and API Gateway + Lambda. Generalized and published by AWS as an official sample (aws-samples, MIT-0).
- **Internal Impact:** Published two more AWS samples, on Organizations governance and an AgentCore agent with governed MCP tools (see Publications). AWS Professional Services MLOps SME program graduate. AWS technical field community ambassador (AI/ML, DevOps, CloudOps, Agentic AI, Serverless); Anthropic Claude Ambassador (2026). Internal tech talks on Terraform testing and GitOps. Associate Speaker Certification (Renewed 2026); Senior Speaker Certification (2026). Security content review qualification (2026). AWS Jam content tester, DevOps and GenAI (2026).

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated an on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – enabling zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for production observability.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized a production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code-quality analysis into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **Published by AWS:** aws-samples/sample-agentcore-harness-agent-terraform (author, MIT-0) – AWS architecture agent on the Bedrock AgentCore harness, all in Terraform (71 resources): 9 MCP tools behind a default-deny Cedar policy, served to Claude Code, Codex and Kiro; promoted to PROD only after an evaluation passes. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Personal projects and opinions are my own and do not represent Amazon or AWS.

Personal Projects

Ledger Sync Jan 2026
React 19, TypeScript, FastAPI, SQLAlchemy, Neon PostgreSQL, Vercel GitHub | Live
◇ Self-hosted personal-finance app: imports .xlsx, .xls and .csv bank statements with SHA-256 transaction IDs so re-imports stay idempotent, then serves 26 analytics pages (Indian fiscal-year tax planning, investments, FIRE, multi-currency) and an AI assistant limited to 15 read-only tool calls. FastAPI on Vercel over Neon PostgreSQL, React on GitHub Pages, with 298 test files across both stacks.

orrery Oct 2026
Agent Skill, Claude Code Plugin, Node.js, ELK, SVG GitHub | Gallery
◇ Agent Skill that turns a YAML spec into animated architecture diagrams with 18,209 official icons (AWS, Azure, Google Cloud, Kubernetes, GitHub, IBM), each cloud pack rebuilt from the vendor's own download and pinned by checksum. ELK auto-layout, a validator whose every finding names the fix, and a review board of 3 reviewer agents; 40 reference diagrams are rendered and checked in CI. Runs in Claude Code, Codex, Cursor, Gemini CLI and Kiro.

Claude Cost Optimizer Mar 2026
Claude Code Plugin, MCP, TypeScript, React, Python GitHub | Live
◇ Claude Code plugin, MCP cost server, CLI grader and web calculator for Claude spend. Rebuilt every estimator on cache-aware billing, correcting a modelled 30-turn Opus session from $3.44 to $0.74, and replaced every vague savings range with a sourced or computed number. Golden-value tests pin the cost math in all three implementations.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

Other Notable Projects: Bedrock Multi-Model MCP (26 text models, 156 of 156 benchmark calls), itr-agent (deterministic tax engine, 151 tests), skillcheck (Agent Skills linter, 284 npm downloads a month), GitScope (Chrome Web Store extension), Kinfolk (collaborative family-graph app on Next.js + Neon).

Open Source Contributions (11 merged upstream)

- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

Cloud & Infrastructure: AWS (Organizations, SCPs, Landing Zone, Control Tower, Transit Gateway, Direct Connect, VPN, Network Firewall, Security Hub, Config, AWS Backup, Systems Manager Patch Manager, IAM, ECS, Lambda, API Gateway, CloudFront, EventBridge, CloudWatch, X-Ray, CodePipeline, CodeDeploy), Terraform, Docker, Ansible
DevOps & Automation: GitHub Actions, CI/CD, GitOps, Blue-Green Deployments, tftest, PyTest, cfn-lint/cfn-guard, SonarQube
MLOps & AI/ML: SageMaker (Pipelines, Endpoints, Custom PSI drift jobs), TensorFlow, PyTorch, Scikit-learn, RAG Pipelines, LLMOps
AI Developer Tools: Claude Code (custom skills, MCP servers, hooks, subagents), Amazon Q, Kiro, GitHub Copilot, MCP Protocol, AWS Bedrock (Nova, Claude, Llama), Amazon Bedrock AgentCore
Languages: Python, HCL, TypeScript, JavaScript, C++, Java, SQL, Bash
Frameworks & Data: React, Next.js, Node.js, Express, FastAPI, MongoDB, PostgreSQL, SQLite